BioMedical QA with KG

Course: DA 331 : Big Data

Tanish Anand — 240150037 **Tumu Sai Seshi Kiran** — 240150038

Abstract

We wanted to find out whether adding facts from a biomedical knowledge graph (KG) to the prompt of a language model helps it answer biomedical research questions. We framed this as a controlled investigation rather than a system-building exercise. On the 500-question development pool of PubMedQA (PQA-L), we gave two small instruction-tuned models (Qwen3-4B-Instruct-2507 and Phi-4-mini-instruct) facts retrieved from either UMLS or PrimeKG, with and without the study abstract, and compared every KG condition with a *control* condition that adds the same number of length-matched but unrelated facts. The abstract raised Qwen3-4B's accuracy from 0.184 to 0.540. KG facts did not help: with the abstract, relevant facts changed accuracy by 0.000 (UMLS) and +0.010 (PrimeKG) compared with control facts, and no such comparison was significant after Holm correction. Without the abstract, both kinds of facts lowered accuracy. Classical classifiers with 18 KG-derived features showed the same picture (all Holm-adjusted $p = 1.00$). A preliminary, AI-assisted audit of 369 retrieved facts found only one that was directly relevant to its question. Our diagnostics point mainly to missing knowledge in the graphs, followed by incomplete entity linking and generic retrieval; whether the models could use genuinely relevant facts remains untested.

1 Introduction

Biomedical questions often depend on specialised facts: which drug treats which disease, which gene is linked to which condition, which symptom belongs to which disorder. Large language models (LLMs) learn many such facts during pre-training, but they can be wrong or incomplete. A common remedy is *retrieval augmentation*: look up relevant information in an external source and add it to the prompt (Lewis et al., 2020). Biomedical knowledge graphs are an attractive source because they store curated relations in a structured, checkable form (Bodenreider, 2004; Chandak et al., 2023), and several systems combine KGs with language models (Yasunaga et al., 2021; Soman et al., 2024).

This led to our central research question:

> *Does structured biomedical knowledge graph augmentation improve biomedical question answering?*

We broke it into four sub-questions: (RQ1) does adding KG facts improve accuracy; (RQ2) does the choice of KG (UMLS vs. PrimeKG) matter; (RQ3) do relevant facts help more than matched *irrelevant* facts; and (RQ4) if not, where does the pipeline break down?

RQ3 is the reason we treat this as a controlled investigation and not as system building. A prompt that contains extra facts differs from one without them in two ways: it contains *knowledge*, and it is *longer and differently shaped*. LLMs are known to react to irrelevant context (Shi et al., 2023), so an accuracy change could come from the extra text alone. We therefore paired every KG condition with a control condition whose facts are real KG facts of the same kind and length but about unrelated concepts. Only the difference between the two can be credited to the *content* of the facts.

Our contributions are: (i) a reproducible pipeline that builds two KGs (UMLS processed with PySpark and a whitelisted PrimeKG), links questions to them and retrieves facts; (ii) a six-condition experiment with matched controls on two LLMs, together with a classical machine-learning (ML) experiment with KG features; (iii) a diagnostic analysis that separates four possible bottlenecks: (A) knowledge, (B) entity linking, (C) retrieval and (D) utilisation. We do not aim for state-of-the-art accuracy; all numbers come from the development pool.

2 Background

Biomedical question answering and PubMedQA. PubMedQA (Jin et al., 2019) turns PubMed research articles into questions. Each question (usually the article title, phrased as a question) comes with the article's abstract *without its conclusion*, and the task is to answer *yes*, *no* or *maybe*, the answer the conclusion supports. Because the conclusion is removed, the model has to reason from the study's background, methods and results. We used the expert-labelled subset PQA-L.

Knowledge graphs. A KG stores knowledge as a network. *Nodes* are concepts (a drug, a disease, a gene) and *edges* are typed relations between two nodes, e.g., "Acebutolol — *contraindicated in* — asthma". The *neighbours* of a node are the nodes it shares an edge with. For a question that mentions two concepts, a *direct relation* is an edge that connects those two concepts; a *neighbour fact* is an edge from one question concept to some other concept the question does not mention. This distinction is central to our analysis (Section 5.2).

UMLS and PrimeKG. The Unified Medical Language System (UMLS) Metathesaurus (Bodenreider, 2004) merges more than a hundred biomedical vocabularies. Each concept has a concept identifier (CUI), many synonyms (MRCONSO), semantic types (MRSTY) and relations contributed by individual source vocabularies (MRREL); MRSAB describes the sources and their licence restrictions. UMLS is very broad but noisy, and most of its relations are hierarchical ("is a") rather than clinical. Access requires a licence, so UMLS-derived text cannot be redistributed. PrimeKG (Chandak et al., 2023) is a public precision-medicine KG that integrates 20 resources into a graph of diseases, drugs, phenotypes, genes/proteins, exposures and other node types, with clinically oriented relations such as *indication*, *contraindication* and *side effect*.

Entity linking. Before a KG can be used, the concepts mentioned in a question must be found and mapped to nodes; this is *entity linking*. "Do inhaled corticosteroids affect perception of dyspnea… in asthma?" should be linked to the nodes for *inhaled corticosteroids*, *dyspnea* and *asthma*. Linking is hard because the same concept has many surface forms ("heart attack", "myocardial infarction") and the same word can mean different concepts ("secondary" in "secondary physis" is not a metastasis).

Split	yes	no	maybe	total
Dev-100 (stratified)	55	34	11	100
Dev-400	221	135	44	400
Dev-500 (dev pool)	**276**	**169**	**55**	**500**

Table 1: PQA-L splits and gold labels. Dev-500 is 55.2% *yes*, 33.8% *no* and 11.0% *maybe*.

We deliberately used a simple, transparent dictionary matcher (Section 5.1) so that every link can be traced.

Metrics. *Accuracy* is the share of questions answered correctly. Because the labels are imbalanced (55% *yes*), we also report *macro-F1*: for each class, precision is the share of predictions of that class that are correct, recall is the share of questions of that class that are found, F1 is their harmonic mean, and macro-F1 is the unweighted mean over the three classes, so the rare *maybe* class counts as much as *yes*. A *confusion matrix* counts each (gold, predicted) pair. Since all conditions answer the *same* questions, we compare them with *paired* statistics: an exact McNemar test (McNemar, 1947) on the questions where two conditions disagree, and a paired bootstrap confidence interval (CI) for the accuracy difference. When several hypotheses are tested together, Holm's procedure (Holm, 1979) adjusts the p-values to control the family-wise error rate.

3 Data and Experimental Setup

Splits. PQA-L has 1,000 questions; we worked only with the 500 that form our *development pool* (Dev-500). We built and debugged the pipeline on a stratified subset of 100 questions (Dev-100; seed 42), then ran the frozen pipeline on all 500 questions. Dev-400 denotes the other 400. All results in this report are on Dev-500 unless stated otherwise. Table 1 gives the label counts. Because 55.2% of the answers are *yes*, a trivial "always yes" classifier already reaches 0.552 accuracy (but a macro-F1 of only about 0.24), which is an important reference point.

Models. Our primary model is Qwen3-4B-Instruct-2507 (Yang et al., 2025), a 4B-parameter instruction-tuned model. To check whether findings depend on one model family, we repeated all LLM runs with Phi-4-mini-instruct (Microsoft et al., 2025), a 3.8B-parameter model from a different family. Both were run with Hugging Face

Transformers on a Colab T4 GPU in fp16 (transformers 5.22.1 was recorded in the run configuration), with greedy decoding and at most 16 new tokens. A strict parser accepted only the labels *yes*, *no* and *maybe*; no output in any run was invalid.

4 Knowledge Graph Construction

UMLS with PySpark. The UMLS files are large text tables (MRCONSO and MRREL are multi-gigabyte files), so we processed them with PySpark (Zaharia et al., 2016) (version 3.5.3, local mode with two workers). Each file is split into byte ranges of 128 MB; every line belongs to exactly one range, each range is filtered independently, and the result is saved as a small Parquet checkpoint, so an interrupted job resumes where it stopped. The filters were:

- **Sources** (MRSAB): current, English sources with licence restriction level 0 (53 sources).
- **Names** (MRCONSO): English strings from those sources whose normalised form matches a word n-gram (1–6 words) of the target text.
- **Relations** (MRREL): only the 11 clinically meaningful relation attributes listed in Appendix B (e.g., *may_treat*, *induces*, *disease_has_finding*); hierarchical relations were not used as facts.
- **Types** (MRSTY): semantic types, used to map concepts to coarse types (disease, drug, phenotype, gene, . . .).

Full vs. question-scoped graph. Keeping every concept of UMLS would be wasteful, because only concepts that can be matched in our texts can ever be retrieved. We therefore built *scoped* graphs: the matched concepts, the whitelisted edges in which they take part, and the names and synonyms of all concepts involved (including the neighbours reached by those edges). The **question-scoped** graph starts from the 500 development questions and is used for the LLM experiments; the **question+abstract-scoped** graph also starts from the abstracts and is used for the ML features and the retrieval-strategy diagnostics. Both are produced by the same Spark pass. PrimeKG is small enough to keep whole: we read its 8.1 million rows in chunks of one million, kept the 13 whitelisted (relation, display relation) types (Appendix B), and, because PrimeKG stores every edge in both directions, kept only the direction whose first node has the listed subject type (e.g., drug → disease for *indication*). Table 2 summarises the resulting graphs.

Graph	Nodes	Edges	Synonyms
PrimeKG (whitelisted)	35,661	370,623	–
UMLS, question-scoped	14,099	23,622	98,357
UMLS, q.+abstract	19,879	46,086	140,379

Table 2: Graphs used in this study. PrimeKG has 12 displayed relation types after whitelisting; UMLS has 11 relation attributes. Only the question-scoped UMLS graph feeds the LLM prompts.

Each relation type has an English sentence template and a *priority* from 1 (most clinically useful, e.g., treatment and contraindication) to 5 (e.g., drug transporters); see Appendix B. For example, the PrimeKG *contraindication* template turns an edge into "Acebutolol is contraindicated in asthma."

5 Entity Linking and Fact Retrieval

Figure 1 shows the pipeline. Linking and retrieval are rule-based and deterministic: nothing in them looks at the meaning of the question or at the abstract, which makes their behaviour easy to audit.

5.1 Entity linking

Normalisation. Both the question and every node name are normalised the same way: Unicode NFKC normalisation, lower-casing, removal of possessive *'s*, punctuation replaced by spaces, and a short list of British-to-American spellings (e.g., *oesophagus* → *esophagus*).

Matching. The linker scans word n-grams from 6 words down to 1, *longest first*, and accepts a span only if it does not overlap an already accepted span, so "inhaled corticosteroids" is preferred over "corticosteroids". A span must match a node name exactly. If it does not, a *plural fallback* removes a trailing *s* from words longer than four characters (*tumours* → *tumour*). Gene symbols are matched only as exact, case-sensitive tokens of at least three characters, so that ordinary words are not read as genes. Spans consisting only of stopwords and overly generic names (*disease*, *syndrome*, *disorder(s)*, *condition*, *injury*) are skipped. When a span matches several nodes, the linker prefers the type order disease > drug > phenotype > exposure > gene, then the lowest node index.

UMLS synonyms vs. PrimeKG names. For PrimeKG the linker matches node names only. For UMLS it also matches all synonyms of a concept, which raises recall but brings in wrong senses: in

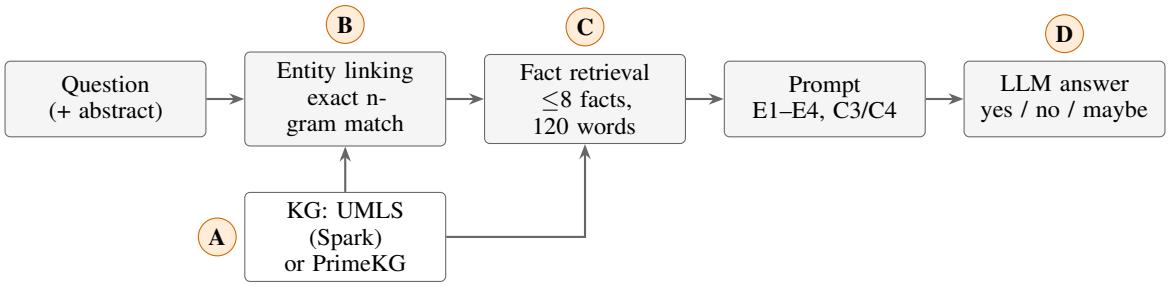


Figure 1: Pipeline of the LLM experiments and the four points where useful knowledge can be lost: (A) the needed relation is not in the KG; (B) the question's concepts are not linked; (C) the retriever does not choose the useful edges; (D) the model does not use the facts it receives.

the audit (Section 9) "plaques" was linked to *senile plaques* and "secondary" to *metastasis*. For the LLM facts only the question is linked; the ML features also link the abstract.

5.2 Fact retrieval

Given the linked question concepts, the retriever builds at most eight sentences:

1. **Direct edges first:** any edge connecting two linked question concepts (at most two per pair).
2. **Candidate neighbours:** for each linked concept, all its whitelisted edges to other concepts.
3. **Ordering:** a concept's candidates are sorted by relation priority, then relation name, then alphabetically by sentence text.
4. **Round robin:** the concepts take turns adding their next candidate, so one concept with hundreds of edges cannot fill the whole budget.
5. **Caps:** at most 2 facts per (concept, relation), 4 per concept, 8 in total and 120 words in total.

The caps keep prompts short, so that the facts do not swamp the question or the abstract; they spread the budget across concepts and relation types; and they make prompts comparable across questions. The price is that for a well-connected concept, alphabetical order decides which of many equally ranked candidates the model sees.

Figure 2 illustrates the result on a real development question. *Dyspnea* and *asthma* are linked, but *inhaled corticosteroids* is not, and the KG has no direct edge between the two linked concepts. All eight retrieved facts are therefore neighbour facts, picked largely by alphabetical order: statements about Abacavir, Acebutolol, Acamprosate, Acetylcysteine, Acefylline, Beclomethasone, acute interstitial pneumonia and acute lung injury. They are true, but none says anything about whether inhaled corticosteroids change how dyspnea is perceived.

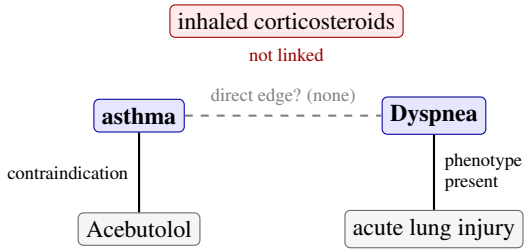


Figure 2: Direct vs. neighbour facts for development question 11555508 with PrimeKG: "Do inhaled corticosteroids affect perception of dyspnea during bronchoconstriction in asthma?" Two concepts are linked (blue); the treatment concept is missed (red). There is no direct asthma–dyspnea edge, so all eight facts are neighbour facts such as the two shown (solid lines).

Cond.	Abstract	KG facts	Control facts
E1	–	–	–
E2	✓	–	–
E3	–	✓	–
C3	–	–	✓
E4	✓	✓	–
C4	✓	–	✓

Table 3: The six conditions; every prompt contains the question. E3, C3, E4 and C4 are run once per KG, giving ten configurations per model. E1→E2 is a positive control: it shows that the setup can detect useful information when it is present.

6 Experimental Conditions and Controls

Table 3 lists the conditions. All share one system prompt, which asks the model to classify the answer as *yes* (affirmative), *no* (negative) or *maybe* (uncertain or conditional) and to reply with the label only (Appendix A). The abstract is introduced as "Abstract (conclusion removed):" and facts appear as a bulleted list under the header "Background facts from a biomedical knowledge graph (they may be incomplete or not relevant to the question):".

KG	Cond.	Acc.	Macro-F1	Pred. yes/no/maybe
–	E1	.184	.161	72 / 8 / 420
–	E2	**.540**	.467	229 / 46 / 225
UMLS	E3	.136	.107	36 / 5 / 459
	C3	.138	.109	36 / 4 / 460
	E4	.532	.456	236 / 45 / 219
	C4	.532	.462	233 / 47 / 220
PrimeKG	E3	.152	.126	52 / 5 / 443
	C3	.150	.124	49 / 5 / 446
	E4	**.544**	.468	234 / 46 / 220
	C4	.534	.458	237 / 45 / 218

Table 4: Qwen3-4B on Dev-500 (gold: 276 yes / 169 no / 55 maybe; always-yes accuracy 0.552). 95% bootstrap CIs for accuracy: E1 [0.150, 0.218], E2 [0.506, 0.580].

KG	Comparison	Diff.	95% CI	Holm p
–	E1→E2	+.356	[.306, .406]	$<.001^{\dagger}$
UMLS	E1→E3	−.048	[−.070, −.026]	.00015
	C3→E3	−.002	[−.010, .006]	1.00
	E2→E4	−.008	[−.024, .008]	1.00
	C4→E4	.000	[−.014, .016]	1.00
PrimeKG	E1→E3	−.032	[−.050, −.014]	.0034
	C3→E3	+.002	[−.004, .008]	1.00
	E2→E4	+.004	[−.014, .022]	1.00
	C4→E4	+.010	[−.002, .022]	.68

Table 5: Qwen3-4B paired accuracy differences on Dev-500, McNemar tests Holm-corrected within each KG family. †Positive control, uncorrected ($p = 3.0 \times 10^{-36}$).

Control facts. For each question that received KG facts, the control set has the *same number* of facts, drawn from the facts retrieved for *other* development questions (seed 42). For each original fact we look for a donor fact with the same relation type, otherwise the same priority, otherwise any relation; among 25 sampled candidates the one closest in word count wins, so control prompts are about as long as KG prompts. Donor facts may not mention any concept linked in the question, nor any neighbour of those concepts; for UMLS we additionally excluded donors that contain question terms as words. Control facts are therefore real, well-formed KG facts that are almost certainly irrelevant.

Questions without facts. UMLS produced facts for 291 of the 500 questions and PrimeKG for 203. For the remaining questions, E3 and C3 are identical to E1, and E4 and C4 to E2. We verified this through the stored prompt hashes: all 209 uncovered UMLS questions and all 297 uncovered PrimeKG questions had prompts byte-identical to the corresponding baseline.

Statistical plan. Before running the 500 questions we fixed three families of comparisons, each Holm-corrected: UMLS (E1→E3, C3→E3, E2→E4, C4→E4), the same four for PrimeKG, and UMLS vs. PrimeKG for each of E3, C3, E4 and C4. The positive control E1→E2 is reported uncorrected. Each comparison uses an exact McNemar test and a paired bootstrap CI (2,000 resamples, seed 42).

7 LLM Results

Table 4 gives the Qwen3-4B results and Figure 3 compares both models.

The abstract helps a lot. With the question alone (E1), Qwen3-4B answers *maybe* 420 times out of 500 and reaches only 0.184 accuracy. Adding the abstract raises accuracy to 0.540 (+0.356, 95% CI [0.306, 0.406], McNemar $p = 3.0 \times 10^{-36}$). This positive control shows that our setup can detect a strong source of useful information. Even so, E2 does not beat the always-yes baseline in accuracy, although its macro-F1 (0.467) is far higher than the baseline's. The E2 confusion matrix explains why: 195 of 276 *yes* questions and 32 of 55 *maybe* questions are answered correctly, but 115 of the 169 *no* questions are answered *maybe*, so only 43 are answered correctly.

KG facts do not help. Figure 4 and Table 5 show the paired comparisons. With the abstract, adding KG facts changed accuracy by -0.008 (UMLS) and $+0.004$ (PrimeKG), and relevant facts differed from control facts by 0.000 and $+0.010$; every CI includes zero and every Holm-adjusted p is at least 0.68. Without the abstract, KG facts *lowered* accuracy (-0.048 for UMLS and -0.032 for PrimeKG, both significant after Holm), but control facts lowered it just as much (C3→E3: -0.002 and $+0.002$). The drop therefore comes from adding facts as such, not from their content.

The choice of KG does not matter. UMLS and PrimeKG differ by at most 0.016 in any condition. The largest difference (E3, -0.016, uncorrected McNemar $p = 0.039$) does not survive Holm correction ($p = 0.154$); the others have Holm $p \geq 0.79$.

Second model. Phi-4-mini-instruct (Table 6) reproduces the main pattern. The abstract helps (+0.294, CI [0.248, 0.338]); without it, the model

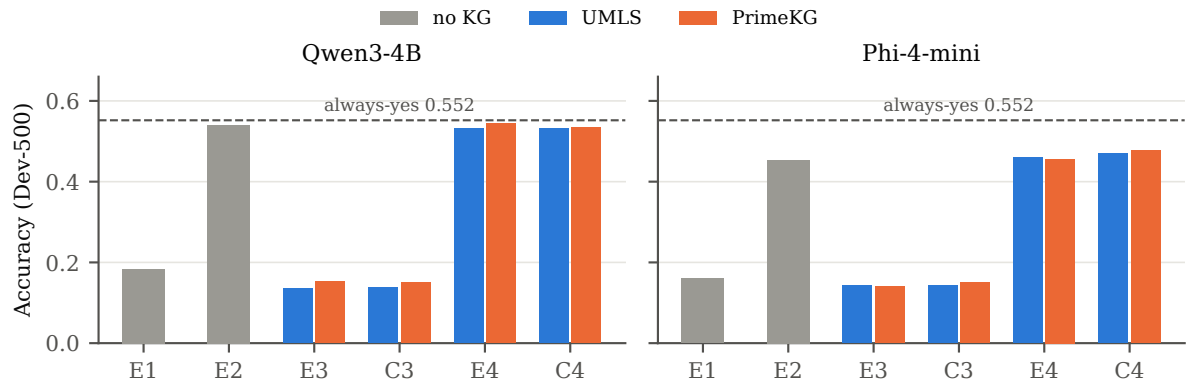


Figure 3: Accuracy on Dev-500 for both models. The abstract (E1→E2) has a large effect; KG facts (E) and matched control facts (C) produce almost identical bars, with or without the abstract. The dashed line is the always-yes baseline.

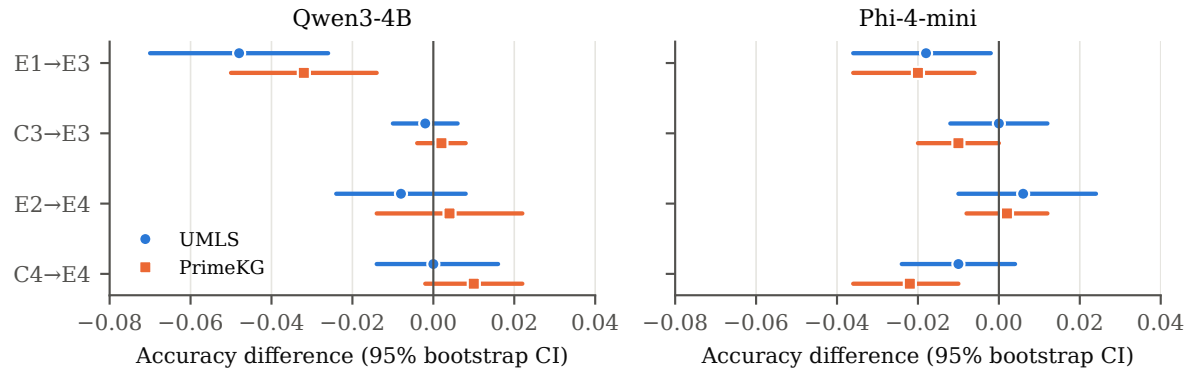


Figure 4: Paired accuracy differences with 95% bootstrap CIs (2,000 resamples). The key comparisons are C3→E3 and C4→E4 (relevant vs. control facts). Only Phi-4-mini's PrimeKG C4→E4 excludes zero, and it is negative: control facts did *better*.

answers *maybe* 438 times out of 500. With the abstract, KG facts change accuracy by +0.006 (UMLS) and +0.002 (PrimeKG), with CIs between about −0.01 and +0.024. One result differs from Qwen: with PrimeKG and the abstract, *control* facts beat relevant facts (−0.022, CI [−0.036, −0.010], Holm $p = 0.004$). An exploratory look at the answer changes suggests a shift in answer tendency rather than knowledge use: on PrimeKG-covered questions, 12 of the 16 answers that change from E2 to C4 move from *maybe* to *yes*, and 13 of the 16 changes turn a wrong answer into a right one, because *yes* is the majority answer. This explanation is post hoc, not a tested hypothesis. All UMLS-vs-PrimeKG comparisons for Phi-4-mini have Holm $p = 1.00$.

Tendency towards *maybe*. The last column of Table 4 shows how Qwen3-4B's answers are distributed. Without the abstract, the model mostly says *maybe*, and adding facts makes this stronger: from E1 to UMLS E3, 41 answers change and 40 of them become *maybe*. On questions that received facts, facts without the abstract break (right → wrong) far more answers than they fix: 29 broken vs. 5 fixed for UMLS E3, and 29 vs. 6 for its control C3.

	UMLS		PrimeKG	
Cond.	Acc.	F1	Acc.	F1
E1	.160	.133	.160	.133
E2	.454	.418	.454	.418
E3	.142	.112	.140	.113
C3	.142	.115	.150	.123
E4	.460	.418	.456	.419
C4	.470	.428	.478	.437

Table 6: Phi-4-mini-instruct on Dev-500 (F1 = macro-F1). E1 and E2 use no KG and are repeated in both columns. Holm-adjusted p (UMLS / PrimeKG): C3→E3 1.00 / 0.25; C4→E4 0.80 / 0.004.

Model	Text	+UMLS	+PrimeKG	UMLS	PrimeKG
LR	**.384**	.386	.364	.291	.233
RF	.337	.339	.334	.338	.288
XGB	.373	.350	.358	.336	.285

Table 7: Classical ML macro-F1 (mean over 15 folds; LR text-only SD 0.042). The last two columns use KG features only. Corresponding accuracies for text only: LR .529, RF .571, XGB .516.

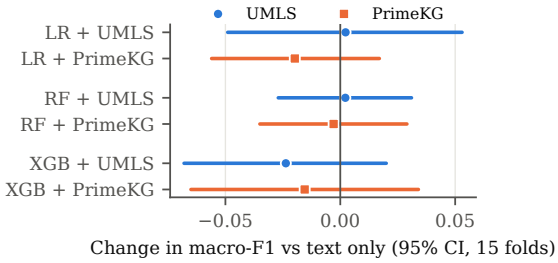


Figure 5: Change in macro-F1 from adding KG features to text (mean and 95% CI over 15 folds). All CIs include zero; all Holm-adjusted $p = 1.00$.

8 Classical Machine Learning

A prompt is only one way to use a KG. We also asked whether KG-derived *features* help a conventional classifier, which can learn how much weight to give them.

Setup. The input text is the question and the abstract (without conclusion). Text features are TF-IDF weights of word unigrams and bigrams (minimum document frequency 2, at most 5,000 terms, sublinear term frequency). The 18 KG features (Appendix C) fall into three groups: *entities* (5; e.g., number of linked question and abstract entities, question entities that reappear in the abstract), *relations* (9; edges among linked entities and counts by relation family: treatment, contraindication, adverse effect, phenotype, gene, other) and *structure* (4; mean and maximum log-degree, shared neighbours and the shortest question–abstract distance, capped at 3). Unlike the LLM facts, these features link both the question and the abstract, so UMLS uses the question+abstract-scoped graph. We trained logistic regression (LR), random forests (RF) and XGBoost (XGB) (Chen and Guestrin, 2016), using class-balanced weights, on five feature sets: text only (A), text + UMLS (B), text + PrimeKG (C), UMLS only and PrimeKG only. Evaluation used 5-fold cross-validation repeated 3 times (15 folds) on Dev-500, with hyperparameters chosen by an inner 3-fold search on macro-F1 (grids in Appendix D). Text + KG was compared with text only using the Nadeau–Bengio corrected resampled t-test (Nadeau and Bengio, 2003), which accounts for overlapping training sets, with Holm correction over the six comparisons.

Results. Table 7 and Figure 5 show that KG features add nothing measurable. The text-only macro-F1 values are 0.384 (LR), 0.337 (RF) and 0.373 (XGB). The changes from adding KG features range from $+0.0023$ (LR + UMLS, CI $[-0.049, 0.053]$, better in 8 of 15 folds) to -0.0237 (XGB + UMLS, CI $[-0.068, 0.020]$, better in only 3 folds), and all six Holm-adjusted p-values are 1.00. KG features alone are clearly weaker than text. Note that RF reaches the highest accuracies (0.571 for text only) mainly because it never predicts *maybe*; its macro-F1 is the lowest. In an LR ablation that drops one feature group at a time, macro-F1 moves by at most 0.016, and KG features receive only a small share of RF and XGB feature importance (RF: 2.8% with UMLS, 2.1% with PrimeKG; XGB: 1.4% and 1.0%), mostly on degree and entity-count features rather than relation features. One likely reason is sparsity: UMLS links entities in 456 questions but finds edges among them in only 33; PrimeKG links entities in 354 questions and finds edges in 48.

9 Diagnostics: Where Does the Pipeline Break?

The null results leave RQ4 open. Figure 1 marks four places where useful knowledge can be lost. We examined the first three; the fourth, utilisation, could not be tested because almost no relevant facts reached the model.

9.1 Retrieval statistics (all 500 questions)

Replaying the retriever with bookkeeping shows that over 99% of the facts are neighbour facts: UMLS supplied 1,275 facts of which only 5 were direct edges between two question concepts, and PrimeKG supplied 909 facts of which 9 were direct. For 71.2% (UMLS) and 83.8% (PrimeKG) of the neighbour facts, the concept–relation group had more candidates than could be kept, so alphabetical order decided which facts the model saw. A covered question had a median of 28 (UMLS) and 104 (PrimeKG) candidate facts, of which only 4.67% and 1.81% fit into the budget; 44 and 30 questions

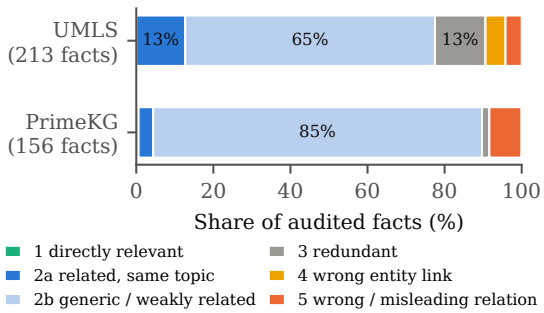


Figure 6: Audit categories of the 369 facts given to the LLM for the 50 audited questions (raw shares; preliminary, AI-assisted, single annotator). Only one fact (PrimeKG, 0.6%) was directly relevant.

reached the eight-fact cap.

9.2 Relevance audit

Design. To judge the facts themselves, we audited a sample of 50 questions. The sample was stratified by coverage and linking behaviour: 10 questions with an ambiguous link (AMB; e.g., a single-word UMLS synonym or a type disagreement between the KGs), 15 with many facts (MANY; at least five in one KG), 15 with few facts (FEW) and 10 with no facts in either KG (NONE), drawn with label quotas (20 yes, 18 no, 12 maybe; seed 42). Every one of the 369 facts shown to the model for these questions (213 UMLS, 156 PrimeKG) was given one of seven categories defined in written guidelines: (1) directly relevant, (2a) indirectly relevant, same topic, (2b) generic or weakly related, (3) redundant with the question, (4) incorrect entity link, (5) incorrect or misleading relationship, (6) cannot determine. The annotator also marked the key concepts of each question and judged every entity link.

Important caveat. This audit is **preliminary and AI-assisted**: the labels were assigned by a single AI annotator (Claude) following our guidelines, they have not been validated by a human or a biomedical expert, and there is no second annotator and hence no agreement statistic. The audit is indicative, not ground truth.

Findings. Figure 6 shows the result. Exactly **one** of the 369 facts was directly relevant: a PrimeKG fact associating the gene CYP2D6 with breast cancer. Most facts were generic (64.8% of UMLS facts, 85.3% of PrimeKG facts) or redundant (13.1% UMLS); 5.2% of UMLS facts came from a wrong entity link and 8.3% of PrimeKG facts stated a misleading relation. Shares re-weighted by each question's inclusion probability (estimated from 1,500 Monte Carlo repetitions of the sampling design) are similar: for PrimeKG 0.8% directly relevant, 87.1% indirectly relevant (2a+2b), 0.8% redundant and 11.3% misleading; for UMLS 0%, 81.9%, 11.9%, 3.2% wrong link and 3.1% misleading. At the level of individual entity links, 81.0% (UMLS) and 83.7% (PrimeKG) were judged correct, 11.1% and 16.3% partially correct, and 6.3% (UMLS only) incorrect.

Key question concepts (146)	UMLS	PrimeKG
Linked	65 (45%)	48 (33%)
Missed	81	98
in KG, usable edges (B)	23	25
in KG, no usable relations (A)	56	–
absent from KG (A)	2	73

Table 8: Key concepts of the 50 audited questions (annotator count). A missed concept that exists with usable edges is a linking failure (B); one without usable relations or absent is a knowledge gap (A).

9.3 Missed concepts: linking vs. knowledge

For every key concept the linker missed, we looked up whether it exists in the KG (Table 8). Only 45% (UMLS) and 33% (PrimeKG) of the 146 key question concepts were linked, and only one question per KG had all its key concepts linked. Of the missed concepts, only 23 (UMLS) and 25 (PrimeKG) existed with usable edges, i.e., could have been found by a better linker. Most were knowledge gaps: in UMLS, 56 missed concepts exist as concepts but have none of the 11 whitelisted clinical relations; in PrimeKG, 73 do not exist as nodes at all. Many key concepts in PubMedQA are study-specific (procedures, outcomes, populations) rather than the drug–disease–gene vocabulary that the KGs cover.

9.4 Alternative retrieval strategies

Could a smarter retriever find better facts in the same graphs? We tried three alternatives on the 50 audited questions (Table 9), using the question+abstract-scoped UMLS graph and linking abstract concepts with the same linker minus single-word synonym links (the main source of wrong senses). R1′ keeps the original candidates and caps but ranks neighbours that also appear in the abstract first; it changed only 4 (UMLS) and 5 (PrimeKG) facts. R2 keeps only edges from a ques-

	Strategy	UMLS	PrimeKG
R1	original retriever	213 (0)	156 (1)
R1′	abstract neighbours first	213 (0)	156 (1)
R2	Q → Q/abstract edges	7 (0)	8 (1)
R3	abstract–abstract edges	0 (0)	1 (0)

Table 9: Retrieval strategies on the 50 audited questions: facts found and, in brackets, questions with a directly relevant fact. Diagnostic runs only; no new LLM predictions.

	UMLS	PrimeKG
Linker as used (50 questions)		
Q–Q direct edge	0	2
Q–Q two-hop path	6	2
Q–abstract direct edge	5	5
Oracle: perfect linking of key concepts		
Questions with ≥2 key concepts	31	23
with a direct key edge	1	4
with a two-hop key path	13	9
Key concept pairs	58	37
with a direct edge	1	5
with a two-hop path	21	12

Table 10: Connectivity between question (Q) and abstract concepts, as linked and under a linking oracle.

tion concept to another question concept or to an abstract concept; it found 7 and 8 edges in total (all redundant for UMLS). R3 looks for edges among abstract concepts and found almost none. No strategy produced a directly relevant fact for a new question; the only question with one (CYP2D6) is the same throughout. A re-run of R1 reproduced the facts the LLM had seen exactly. Re-ranking cannot add relations that are not in the graph.

9.5 Connectivity oracle

Finally, we asked what would happen with perfect linking. The *oracle* adds every missed key concept that exists in the KG with edges and checks whether two key concepts of a question are connected (Table 10). Even then, a direct edge between key concepts exists in only 1 (UMLS) and 4 (PrimeKG) of the 50 questions; two-hop paths are more common (13 and 9 questions). Connectivity is not relevance, however: when the AI annotator labelled the oracle edges, only one was directly relevant (again CYP2D6); the others were redundant, merely related or misleading.

9.6 Interpretation

Taken together, the evidence suggests that the main bottleneck is **knowledge (A)**: the relations a PubMedQA question needs are rarely in either graph, even with perfect linking. **Linking (B)** is a second, smaller problem (about a quarter of missed concepts were findable), and UMLS synonyms add wrong senses. **Retrieval (C)** makes things worse by filling the budget with generic neighbour facts chosen alphabetically, but better ranking did not find relevant facts. **Utilisation (D)** remains untested: because almost no relevant facts reached the models, our results do *not* show that these LLMs cannot use KG facts. All diagnostic evidence rests on the preliminary AI-assisted audit described above.

10 Discussion

The answer to our central question, for this setup, is no: adding UMLS or PrimeKG facts to the prompt did not improve PubMedQA accuracy for either model, and KG features did not improve classical classifiers. Without the matched controls we might have misread the results: the significant drop from E1 to E3 looks like "KG facts hurt", and Phi-4-mini's PrimeKG E2→C4 gain could look like "facts help". The controls show that both are effects of adding text, not of the knowledge in it.

The contrast with the abstract is instructive. The abstract contains study-specific evidence (design, population, results) that bears directly on the answer, and it raised accuracy by 29–36 points. KG facts are general background knowledge about the concepts in the question. PubMedQA asks what *a particular study* found, so general facts seldom decide the answer. This matches the observation by Xiong et al. (2024) that the value of retrieval in medical QA depends strongly on the corpus being retrieved from. It also suggests that reported KG gains elsewhere (Soman et al., 2024) may depend on question types that ask about the relations a KG stores.

11 Limitations

- **Development pool only.** All results are on the 500 development questions. The pipeline was designed on Dev-100, which is part of Dev-500.
- **Two small models, one prompt.** We tested 3.8B- and 4B-parameter instruction-tuned models with one fixed prompt and greedy decoding. Larger models, other prompts or chain-of-thought reasoning might use facts differently.
- **Simple linking and retrieval.** Exact dictionary matching misses paraphrases and cannot disambiguate senses; a learned entity linker might link

more concepts. The retriever ignores question meaning and uses alphabetical tie-breaking. Our strategy analysis suggests this is not the main bottleneck, but we did not test semantic retrieval.

- **Preliminary audit.** The relevance audit covers 50 questions and was labelled by a single AI annotator without human or expert validation and without an agreement measure. Its category shares are indicative only.
- **Utilisation untested.** Since almost no relevant facts were retrieved, we could not test whether the models would use relevant facts if they received them; an experiment with hand-written or oracle facts would be needed.
- **Statistical power.** With 500 questions, differences of about one to two accuracy points cannot be ruled out; our CIs show the range of effects compatible with the data. The Phi-4-mini answer-change explanation is exploratory.

12 Conclusion

We set out to test, under controlled conditions, whether structured biomedical KG facts improve biomedical question answering. On PubMedQA's development pool, they did not: neither UMLS nor PrimeKG facts improved two small LLMs beyond matched irrelevant facts, and KG features did not improve classical classifiers, while the study abstract helped substantially. Our diagnostics suggest that the needed knowledge is rarely present in the graphs, that linking finds under half of the key concepts, and that retrieval returns generic neighbour facts; whether models could use relevant facts is still open. Our contribution is not a better system but a controlled investigation with matched controls and a bottleneck analysis that explains a null result. Natural next steps are an oracle-fact condition to test utilisation, a learned entity linker, and human expert validation of the audit.

Data Licensing

PubMedQA and PrimeKG are publicly released datasets, used under their licences. UMLS is licensed by the U.S. National Library of Medicine and was used under that licence.

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A Prompts

All conditions use the same system prompt (version "e1-v1"):

```
You are a careful biomedical
expert.  You will be given one
biomedical research question.
Classify the answer to the
question as exactly one of these
labels:
- yes:  the answer is
affirmative
- no:  the answer is negative
- maybe:  the answer is
uncertain or conditional
Reply with only the label (yes,
no, or maybe), in lowercase,
with no explanation.
```

The user message (context version "kg-v1") consists of up to three blocks separated by blank lines, in this order: `Abstract (conclusion removed): ...` (E2, E4, C4); the fact header followed by one `- fact` line per fact (E3, E4, C3, C4); and `Question: ...\nAnswer (yes, no, or maybe):`. Messages are formatted with each model's own chat template.

B Relation Whitelists and Templates

PrimeKG whitelist (subject type, priority): indication, contraindication, off-label use (drug, 1); side effect (drug, 2); phenotype present, phenotype absent (disease, 3); linked to (exposure, 3); associated with, disease–protein (disease, 4); target (drug, 4); enzyme, transporter, carrier (drug, 5); associated with, phenotype–protein (phenotype, 5). Each type has a one-sentence template, e.g., "{s} is contraindicated in {o}."
UMLS relation attributes (priority, source vocabulary): may_treat, contraindicated_with_disease, may_prevent (1, MED-RT); may_diagnose, induces (2, MED-RT); has_manifestation (3, OMIM); disease_has_finding, disease_may_have_finding, disease_excludes_finding (3, NCI); disease_has_associated_gene, disease_may_have_associated_disease (4, NCI). Each also has a sentence template (e.g., "{s} may treat {o}.").

C KG Features for Classical ML

Feature	Meaning
Entities (5)	
n_q_entities	linked entities in the question
n_abs_entities	linked entities in the abstract
n_unique_entities	distinct linked entities
any_entity	at least one entity linked
n_q_in_abs	question entities found again in the abstract
Relations (9)	
n_edges_within	edges among all linked entities
n_edges_qq	question–question edges
n_edges_qa	question–abstract edges
rel_treat, rel_contra, rel_adverse, rel_phenotype, rel_gene, rel_other	edge counts by relation family
Structure (4)	
mean_log_degree, max_log_degree	log-degree of linked entities
n_shared_neighbours	neighbours shared by linked entities
qa_min_dist	shortest question–abstract distance (capped at 3)

Table 11: The 18 KG features, computed separately for each KG.

D Hyperparameters

LR: $C \in \{0.1, 1, 10\}$, balanced class weights. **RF**: 300 trees, balanced-subsample class weights, maximum depth $\in \{\text{None}, 20\}$, minimum samples per leaf $\in \{1, 3\}$. **XGB**: 200 trees, learning rate 0.1, histogram tree method, maximum depth $\in \{3, 6\}$, L2 regularisation $\lambda \in \{1, 10\}$, balanced sample weights.